

2026/05/19

Quiz 5 Class 2 Solutions

Computer Security Fundamentals Spring 2026

Instructions: Answer the questions, **closed-book, closed-notes**. Time: 10 minutes

1. (**F**) For different threads in a multi-threading environment, each thread has its own virtual memory space and registers. (True/False, 2 pts)

2. (**C**) Which of the following is a classic example of a TOCTOU (Time-of-Check to Time-of-Use) race condition? (2 pts)

- A. A program crashes because it was not allocated with enough memory.
- B. A forged DNS response tries to get to the original sender before the real one arrives.
- C. A process opens a file without knowing the path was changed after verification.
- D. A user cannot log in because the password was not inputted in time.

3. (**A**) Normally, an operating system kernel has the following procedures: (2 pts)

- A. Read from userspace, execute instructions, write to userspace, return to userspace
- B. Write to userspace, execute instructions, read from userspace, return to userspace
- C. Read from userspace, execute instructions, write to userspace
- D. Write to userspace, execute instructions, read from userspace

4. (Short Answer, 4 pts) Perform taint analysis for the source code below.

```
printf(untainted char *s);
tainted char *get_env_var();

void logger(int level) {
    char *buf;
    char *env = get_env_var();

    if (level == 5) {
        buf = env;
    } else {
        buf = "System Boot";
    }

    if (level < 3) {
        printf(buf);
    }
}

// Main call
logger(2);
```

Solution:

Use the names `ae` and `ab` for the local variables `env` and `buf` in the `logger()` function, and `p` for the parameter level.

Constraints:

$p-1 \geq \text{untainted}$ (from the call `logger(2)`)

$ae \geq \text{tainted}$ (from `get_env_var()`)

Path Sensitivity:

For `logger(2)`, the parameter level is 2.

`level == 5` is False, so the program executes the else branch: `buf = "System Boot"`.

$ab \geq \text{untainted}$

`level < 3` is True, so the program executes `printf(buf)`.

Sink Constraint:

$\text{untainted} \geq ab$

Final Flow: $\text{untainted} \geq \text{untainted}$. This is a legal flow.